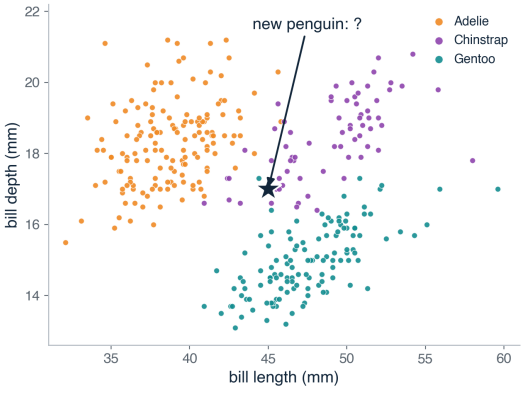
LDA and QDA

MATH 4432 Statistical Machine Learning · Tutorial T06

Department of Mathematics, HKUST

Fall 2026

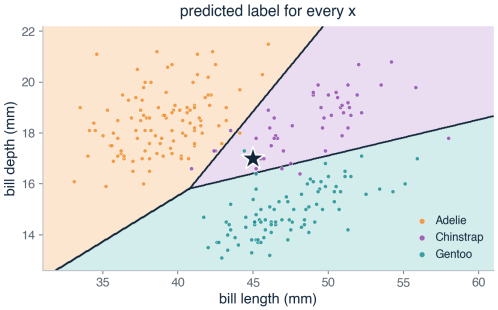
What is classification?

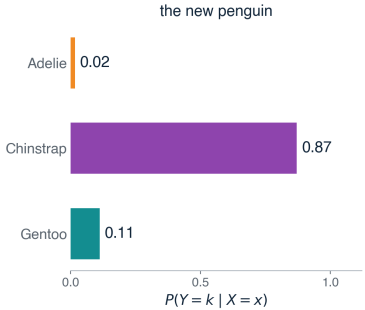


- **observation** i
  features x_i: bill length, bill depth
  label y_i: species
- **training data**
  342 labelled penguins
- **goal**
  predict y for a new x

regression: y is a number
classification: y is a category

What a classifier gives you





probabilities

$P(Y = k \mid X = x)$, one per class

a label

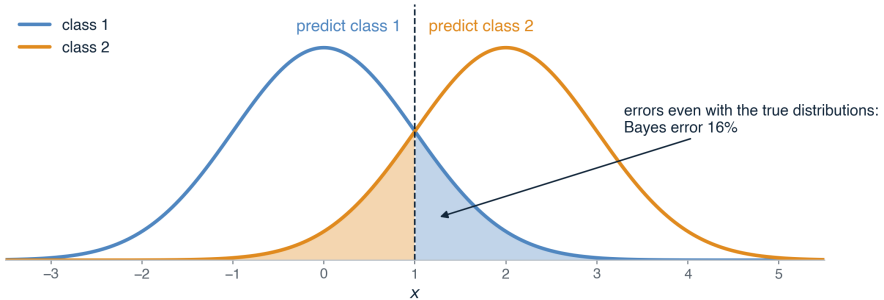
the most probable class

a decision boundary

where the label changes

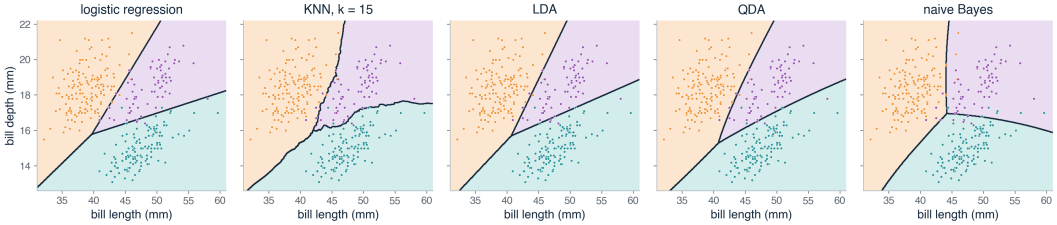
The best possible classifier

$$\hat{y}(x) = \arg\max_k P(Y = k \mid X = x)$$ the **Bayes classifier**



in practice $P(Y = k \mid X = x)$ is unknown: every classifier **estimates** it

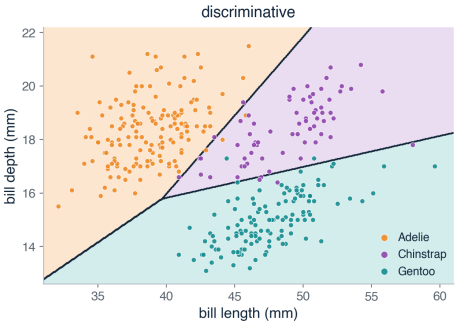
Why so many classifiers?

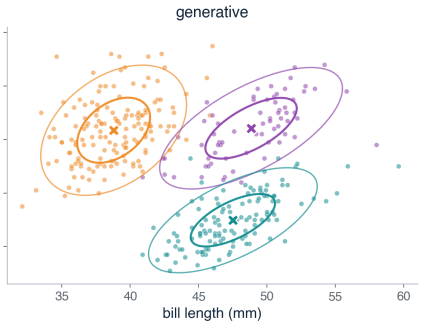


same 342 penguins, five classifiers, five different boundaries

each estimates $P(Y = k \mid X = x)$ under a different **assumption**

Two routes to $P(Y = k \mid X = x)$





discriminative: learn the boundary

model $P(Y = k \mid X = x)$ directly

generative: learn each class

model $f_k(x)$ and π_k, then Bayes' theorem

Bayes' theorem

$$P(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

- **posterior** $P(Y = k \mid X = x)$: probability of class k after seeing x
- **prior** $\pi_k = P(Y = k)$: how common class k is
- **likelihood** $f_k(x)$: how typical x is for class k (the density of X in class k)
- **denominator**: the same for every class; makes the probabilities sum to 1

posterior $\propto$ prior $\times$ likelihood

Parametric vs nonparametric

parametric

a formula with a fixed number of parameters

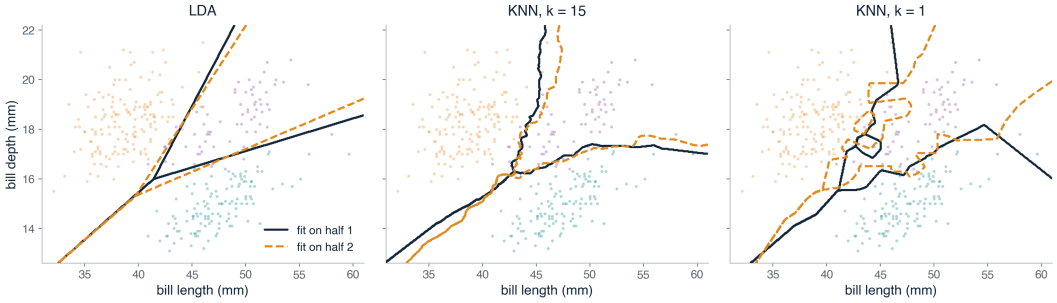
- logistic regression: 6 numbers
  LDA: 11 numbers whatever n is
- fitting = estimating those numbers
- \+ stable with little data
- – biased if the formula is wrong

nonparametric

no fixed formula: the data are the model

- KNN: keeps all 342 penguins
- more data ⇒ more detail
- \+ very flexible, few assumptions
- – needs more data, varies more

Flexible classifiers vary more



fits on two random halves disagree on 0.9%, 2.3%, 5.6% of the penguins

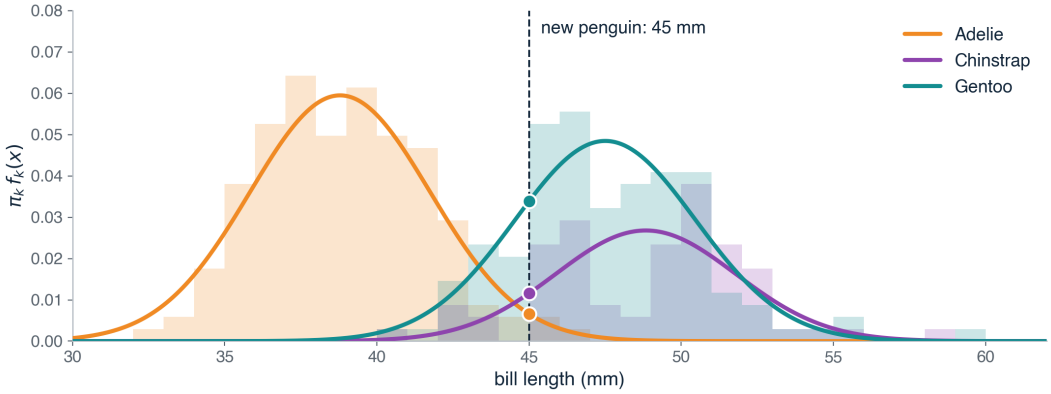
more flexible ⇒ follows the sample more closely ⇒ lower **bias**, higher **variance** (T02)

A map of classifiers

	discriminative model $P(Y \mid X)$	**generative** model $f_k(x)$ and π_k
parametric fixed number of parameters	**logistic regression** log odds linear in x	**LDA** Gaussian classes, shared Σ **QDA** Gaussian classes, own Σ_k **naive Bayes** independent features
nonparametric flexibility grows with n	**KNN** nearby $x \Rightarrow$ similar class	**kernel density classifier** f_k estimated by smoothing

each method is a different assumption about $P(Y = k \mid X = x)$ or about $f_k(x)$

Describe each class, then compare



342 penguins, three species · a 45 mm bill: Gentoo 0.65, Chinstrap 0.22, Adelie 0.13

The generative recipe

observations (x_i, y_i), $i = 1, \dots, n$, $x_i \in \mathbb{R}^p$, $y_i \in \{1, \dots, K\}$

1. **prior** $\pi_k = P(Y = k)$ — class proportions
2. **class-conditional density** $f_k(x)$ — a model for X within class k
3. **Bayes' theorem** $p_k(x) = P(Y = k \mid X = x) = \dfrac{\pi_k f_k(x)}{\sum_l \pi_l f_l(x)}$
4. **classify** $\arg\max_k \ \pi_k f_k(x)$ — the denominator is the same for every k

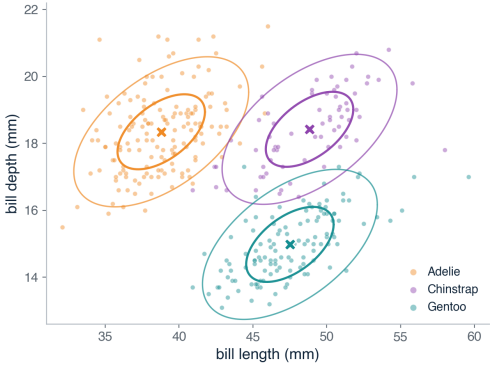
LDA, QDA and naive Bayes differ only in step 2

LDA: Gaussian classes, shared covariance

$$X \mid Y = k \sim N_p(\mu_k, \Sigma)$$

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}$$

- μ_k: centre of class k
- Σ: the same spread and orientation for every class



one ellipse shape, three centres

The LDA discriminant function

$$\log\left(\pi_k f_k(x)\right) = \underbrace{-\tfrac{1}{2}x^\top\Sigma^{-1}x - \tfrac{1}{2}\log|\Sigma| - \tfrac{p}{2}\log 2\pi}_{\text{same for every } k} + x^\top\Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top\Sigma^{-1}\mu_k + \log\pi_k$$

$$\delta_k(x) = x^\top\Sigma^{-1}\mu_k - \tfrac{1}{2}\mu_k^\top\Sigma^{-1}\mu_k + \log\pi_k$$

- δ_k is **linear** in x $\Rightarrow$ boundary $\{x : \delta_k(x) = \delta_l(x)\}$ is a line (hyperplane)
- posterior: $p_k(x) = e^{\delta_k(x)} / \sum_l e^{\delta_l(x)}$

Estimating the parameters

$$\hat{\pi}_k = \frac{n_k}{n}, \qquad \hat{\mu}_k = \frac{1}{n_k} \sum_{i\,:\,y_i = k} x_i$$

$$\hat{\Sigma} = \frac{1}{n - K} \sum_{k=1}^{K} \sum_{i\,:\,y_i = k} (x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^{\mathsf{T}}$$

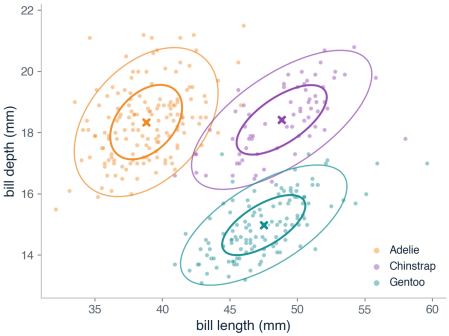
- $\hat{\Sigma}$: **pooled** within-class covariance each class centred at its own mean
- plug in $\Rightarrow \hat{\delta}_k(x)$ penguins: $\hat{\Sigma} = \begin{pmatrix} 8.76 & 1.75 \\ 1.75 & 1.26 \end{pmatrix}$
- ISL and scikit-learn divide by $n - K$; maximum likelihood divides by n

QDA: each class has its own covariance

$$X \mid Y = k \sim N_p(\mu_k, \Sigma_k)$$

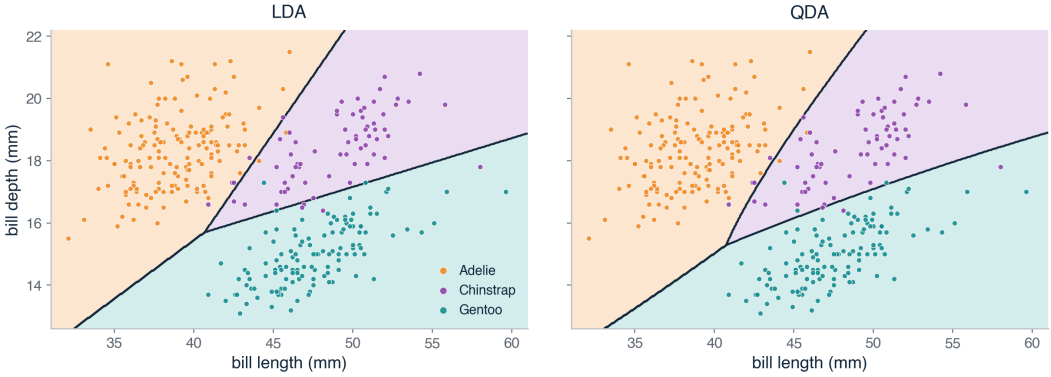
$$\delta_k(x) = -\frac{1}{2}(x-\mu_k)^T\Sigma_k^{-1}(x-\mu_k) - \frac{1}{2}\log|\Sigma_k| + \log \pi_k$$

- $x^T\Sigma_k^{-1}x$ differs across k: no longer cancels ⇒ **quadratic** boundaries
- $\hat{\Sigma}_k = \dfrac{1}{n_k - 1}\displaystyle\sum_{i:\, y_i = k}(x_i - \hat{\mu}_k)(x_i - \hat{\mu}_k)^T$



within-class correlation 0.39, 0.65, 0.64

LDA vs QDA on the penguins



LDA: straight lines · QDA: curves, mild here because the three covariances are similar

The price of flexibility

covariance parameters, $K = 3$

p	LDA $\frac{p(p+1)}{2}$	QDA $K\frac{p(p+1)}{2}$
2	3	9
4	10	30
13	91	273
50	1275	3825

- QDA needs $n_k > p$ in every class, or $\hat{\Sigma}_k$ is singular
- LDA: lower variance, but **biased** if the covariances differ
- QDA: lower bias, higher **variance** $\Rightarrow$ needs more data

1 · fit with scikit-learn

```
from sklearn.discriminant_analysis import (LinearDiscriminantAnalysis,
                                           QuadraticDiscriminantAnalysis)
peng = pd.read_csv("data/penguins.csv").dropna(subset=["bill_length_mm", "bill_depth_mm"])
X = peng[["bill_length_mm", "bill_depth_mm"]].to_numpy()
y = peng["species"].to_numpy()
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.3, stratify=y, random_state=0)
lda = LinearDiscriminantAnalysis().fit(X_tr, y_tr)
qda = QuadraticDiscriminantAnalysis().fit(X_tr, y_tr)
print((lda.predict(X_te) != y_te).sum(), (qda.predict(X_te) != y_te).sum())
print(lda.predict_proba([[45, 17]]).round(3), qda.predict_proba([[45, 17]]).round(3))
```

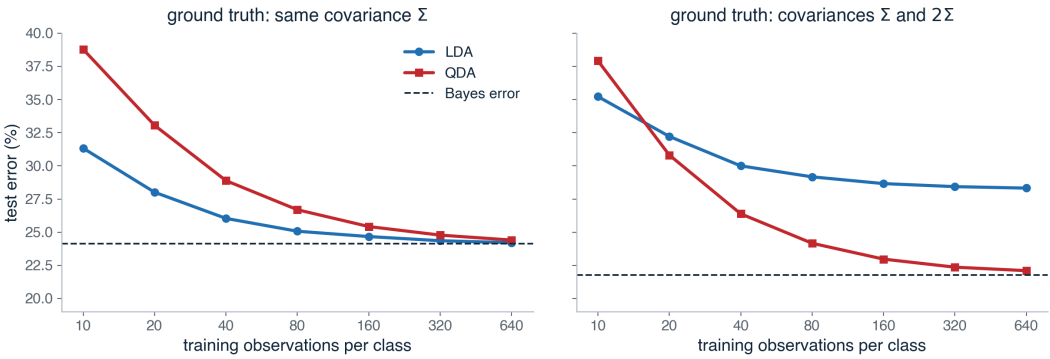
```
4 2                                          # test errors, out of 103
[[0.017 0.871 0.112]] [[0.034 0.954 0.012]]  # Adelie, Chinstrap, Gentoo
```

2 · LDA and QDA by hand

```
def fit_da(X, y, alpha=0.0):              # alpha = 0: LDA,  alpha = 1: QDA
    classes, n, K = np.unique(y), len(y), len(np.unique(y))
    prior = np.array([np.mean(y == k) for k in classes])
    mu = np.array([X[y == k].mean(axis=0) for k in classes])
    S_k = [np.cov(X[y == k].T) for k in classes]                # divides by n_k - 1
    S_pool = sum((np.sum(y == k) - 1) * S for k, S in zip(classes, S_k)) / (n - K)
    S = [alpha * Sk + (1 - alpha) * S_pool for Sk in S_k]
    return classes, prior, mu, S
def predict_da(model, X):
    classes, prior, mu, S = model
    scores = [-0.5 * np.einsum("ij,ij->i", (X - m) @ np.linalg.inv(Sk), X - m)
              - 0.5 * np.linalg.slogdet(Sk)[1] + np.log(p)
              for p, m, Sk in zip(prior, mu, S)]
    return classes[np.argmax(scores, axis=0)]
```

same predictions as scikit-learn on all 103 test penguins, for LDA and for QDA

3 · ground truth known



two Gaussian classes, $p = 5$, $\Sigma_{ij} = 0.5^{|i-j|}$, $\|\mu_1 - \mu_2\| = 2$ · 200 repetitions

What each method assumes

	$\log \frac{P(Y = k \mid X = x)}{P(Y = K \mid X = x)}$	assumption	penguins test error
logistic regression	$\beta_{k0} + \sum_j \beta_{kj} x_j$	log odds linear	3.9%
LDA	$a_k + \sum_j b_{kj} x_j$	Gaussian, shared Σ	4.1%
QDA	$a_k + \sum_j b_{kj} x_j + \sum_j \sum_l c_{kjl} x_j x_l$	Gaussian, own Σ_k	3.7%
naive Bayes	$a_k + \sum_j g_{kj}(x_j)$	independent features	6.7%
KNN, $k = 15$	any shape	nearby x, similar class	4.1%

ISL §4.5 · LDA and logistic: same form, different fit · error: mean of 200 random splits

Take away

- a classifier estimates $P(Y = k \mid X = x)$; the best possible rule picks the most probable class
- discriminative: model $P(Y \mid X)$ directly; generative: model each class, then Bayes' theorem
- every method is an assumption: more flexible $\Rightarrow$ lower bias, higher variance
- LDA: Gaussian classes, shared $\Sigma \Rightarrow$ linear boundaries; QDA: own $\Sigma_k \Rightarrow$ quadratic boundaries
- which one wins depends on the truth and on n: compare on held-out data

References

- James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning*, §2.2.3, §4.4–4.5
- Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, §4.3
- Fisher (1936), The use of multiple measurements in taxonomic problems — the first LDA
- Horst, Hill & Gorman (2020), `palmerpenguins` — the data
- `T06_LDA_QDA.ipynb` — every number on these slides, and more